Lecture 1

Groups acting

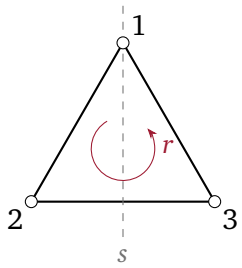
Objectives. *The course asks what the structure of a finite group is, and how its linear representations reflect it. This lecture sets up its basic tool, a group* acting *on a set. From its main result, orbit–stabiliser, follow Lagrange's theorem, the class equation, orbit counting and Cauchy's theorem.*

By the end of this lecture you will be able to:

- Check that a rule is an action, and find its orbits, stabilisers and kernel.
- Prove orbit–stabiliser, and deduce Lagrange's theorem and the class equation from it.
- Find the classes and centralisers of a small group, and write its class equation.
- Count orbits by averaging fixed points; prove Cauchy's theorem.

1.1 Groups, subgroups, homomorphisms

Symmetries of an equilateral triangle compose and can be undone.



Definition 1.1 (Group). A *group* is a set G with a product $G \times G \to G$, $(g,h) \mapsto gh$, such that

- $(gh)k = g(hk)$ for all $g,h,k \in G$;
- there is an $e \in G$ with $eg = ge = g$ for all g;
- each g has a $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

The *order* $|G|$ is the number of elements. G is *abelian* if $gh = hg$ for all g,h.

If e, e' are identities, then

$$e = ee' = e'.$$

If h, h' are inverses of g, then

$$h = h(gh') = (hg)h' = h'.$$

Identities and inverses are therefore unique. Also $(gh)^{-1} = h^{-1}g^{-1}$.

Example 1.2 (Groups).

- The nth roots of unity in $\mathbb{C}^\times$: the *cyclic group* C_n.
- The bijections of a set X to itself, under composition: the *symmetric group* S_X. For $X = \{1, \dots, n\}$ it is S_n, of order $n!$. A permutation $\sigma \in S_n$ is written in cycles: the cycle $(a_1\, a_2\, \dots\, a_k)$ sends $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes everything else. Permutations compose as functions, right to left: $(\sigma\tau)(i) = \sigma(\tau(i))$. For instance
$$(1\,2)(1\,3) = (1\,3\,2).$$
- The *dihedral group* D_n ($n \geq 3$): the orthogonal maps of $\mathbb{R}^2$ that carry a regular n-gon, centred at 0, to itself. With r the rotation by $2\pi/n$ and s the reflection in the line through vertex 1 (drawn above for $n = 3$),
$$r^n = e, \qquad s^2 = e, \qquad srs^{-1} = r^{-1}.$$
Example 1.13 shows that $|D_n| = 2n$.

Definition 1.3 (Subgroup). A *subgroup* $H \leq G$ is a subset that contains e and is closed under products and inverses. $\langle S \rangle$ is the smallest subgroup containing $S \subseteq G$.

Definition 1.4 (Homomorphism). A *homomorphism* is a map $\varphi\colon G \to H$ with $\varphi(gh) = \varphi(g)\varphi(h)$ for all g, h. Its *kernel* is $\ker\varphi = \{g : \varphi(g) = e\}$ and its *image* $\operatorname{im}\varphi$. A bijective homomorphism is an *isomorphism*, written $G \cong H$.

From $\varphi(e) = \varphi(e)\varphi(e)$ we get $\varphi(e) = e$, and then $\varphi(g^{-1}) = \varphi(g)^{-1}$; so $\ker\varphi$ and $\operatorname{im}\varphi$ are subgroups. Moreover $\varphi(g) = \varphi(h)$ if and only if $g^{-1}h \in \ker\varphi$, so φ is injective exactly when $\ker\varphi = \mathbf{1} = \{e\}$.

Example 1.5 ($D_3 \cong S_3$). Each $g \in D_3$ permutes the vertices, and the permutation of gh is that of g after that of h: a homomorphism $D_3 \to S_3$. It is injective, since a linear map fixing vertices 1 and 2, two independent vectors, is the identity. From the figure,

g	e	r	r^2	s	rs	r^2s
permutation	e	$(1\,2\,3)$	$(1\,3\,2)$	$(2\,3)$	$(1\,2)$	$(1\,3)$

which is all of S_3. So $D_3 \cong S_3$, the running example.

1.2 Actions

Definition 1.6 (Action). A (left) *action* of G on a set X is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, with
$$e \cdot x = x,$$
$$g \cdot (h \cdot x) = (gh) \cdot x$$
for all $g, h \in G$ and $x \in X$; X is then a *G-set*. The *orbit* of x is $G \cdot x = \{g \cdot x : g \in G\}$, its *stabiliser* is $G_x = \{g : g \cdot x = x\}$. The action is *transitive* if there is one orbit.

G_x is a subgroup, by the axioms: $g \cdot x = x$ gives $x = g^{-1} \cdot x$. The orbits partition X, as "$y = g \cdot x$ for some g" is an equivalence relation (take e, g^{-1}, or a product).

Proposition 1.7 (Actions are homomorphisms to S_X). *An action of G on X gives bijections $\rho(g)\colon x \mapsto g \cdot x$ and a homomorphism $\rho\colon G \to S_X$; conversely, a homomorphism ρ gives the action $g \cdot x = \rho(g)(x)$. The* kernel *of the action is* $\ker\rho = \bigcap_x G_x$.

Proof. The axioms say exactly $\rho(e) = \mathrm{id}$ and $\rho(g)\rho(h) = \rho(gh)$; so $\rho(g^{-1})$ inverts $\rho(g)$. And $\rho(g) = \mathrm{id}$ exactly when g fixes every x. ∎

The action is *faithful* if its kernel is $\mathbf{1}$; then $G \cong \operatorname{im}\rho \leq S_X$, as for D_3 on the vertices (Example 1.5).

Example 1.8 (Cosets)**.** For $H \leq G$, the *left coset* of x is $xH = \{xh : h \in H\}$. They form the set G/H, whose size is the *index* $[G : H]$. G acts on G/H by $g \cdot xH = gxH$, transitively; gxH is the image of the set xH under $y \mapsto gy$, so it depends only on xH.

Two cosets agree, $xH = yH$, exactly when $x^{-1}y \in H$. If $xH = yH$, then $y \in xH$. Conversely, if $y = xh$ with $h \in H$, then $hH = H$ and

$$yH = x(hH) = xH.$$

Example 1.9 (Guard: G/H is a G-set, not a group)**.** The rule $xH \cdot yH = xyH$ is not well defined in general. In S_3 take $H = \{e, (1\,2)\}$. Then $eH = (1\,2)H$, but with $y = (1\,3)$,

$$\begin{aligned} e\,yH &= \{(1\,3),\ (1\,2\,3)\}, \\ (1\,2)\,yH &= (1\,3\,2)H \\ &= \{(1\,3\,2),\ (2\,3)\}. \end{aligned}$$

Lecture 2 shows that the rule is well defined when H is normal, and then G/H is a group; Lecture 8 uses G/H for every H, and needs only the action.

1.3 Orbit–stabiliser and Lagrange

The cosets of a stabiliser are a model of every orbit.

Theorem 1.10 (Orbit–stabiliser)**.** *Let G act on X and let $x \in X$. The map*

$$G/G_x \to G \cdot x, \qquad gG_x \mapsto g \cdot x,$$

is well defined and bijective, and it carries the action of G on cosets to the action on the orbit. In particular $|G \cdot x| = [G : G_x]$.

Proof. For $g, h \in G$,

$$\begin{aligned} gG_x = hG_x &\iff g^{-1}h \in G_x \\ &\iff g^{-1}h \cdot x = x \\ &\iff h \cdot x = g \cdot x. \end{aligned}$$

Read from left to right, the chain says the map is well defined; read from right to left, that it is injective. It is onto $G \cdot x$ by the definition of the orbit. Finally $k \cdot gG_x = kgG_x$ goes to $kg \cdot x = k \cdot (g \cdot x)$. ∎

Corollary 1.11 (Lagrange)**.** *If G is finite and $H \leq G$, then $|G| = [G : H]\ |H|$. In particular $|H|$ divides $|G|$, and, for any action of G, $|G \cdot x| = |G|\,/|G_x|$.*

Proof. The action. Let H act on G by $h \cdot g = gh^{-1}$. The identity fixes every g, and

$$h \cdot (h' \cdot g) = gh'^{-1}h^{-1} = g(hh')^{-1} = (hh') \cdot g.$$

The orbits. The orbit of g is gH. Its stabiliser is $\mathbf{1}$, since $gh^{-1} = g$ forces $h = e$. By Theorem 1.10, each orbit has $|H|$ elements.

The count. The orbits partition G into its $[G : H]$ left cosets, so $|G| = [G : H]\ |H|$. Apply this formula with $H = G_x$, then use orbit–stabiliser, to obtain the final claim. ∎

In a finite group the powers of g repeat, and $g^i = g^j$ with $i < j$ gives $g^{j-i} = e$. So the *order* $o(g)$, the least $k \geq 1$ with $g^k = e$, exists. The same argument shows that $e, g, \ldots, g^{o(g)-1}$ are distinct, and they form $\langle g \rangle$, since $g^{o(g)} = e$; so $|\langle g \rangle| = o(g)$.

Corollary 1.12. *In a finite group, $o(g)$ divides $|G|$. A group of prime order p is cyclic, generated by any $g \neq e$.*

Proof. Apply Corollary 1.11 to $\langle g \rangle$. If $|G| = p$ and $g \neq e$, then $o(g)$ divides p and is not 1, so $\langle g \rangle = G$. ∎

Example 1.13 ($|D_n| = 2n$)**.** D_n acts on the n vertices, transitively: r^k takes vertex 1 to vertex $k+1$. A symmetry maps vertices to vertices and keeps distances, so if it fixes vertex 1 it sends vertex 2 to a neighbour of vertex 1, vertex 2 or vertex n; and being linear, it is determined by the images of the independent vectors at vertices 1 and 2. So the stabiliser of vertex 1 is $\{e, s\}$, and by Corollary 1.11, $|D_n| = 2n$. The $2n$ elements r^k and $r^k s$ $(0 \leq k < n)$ are distinct ($o(r) = n$, and $\det r^k s = -1$), so

$$D_n = \{e, r, \ldots, r^{n-1},\ s, rs, \ldots, r^{n-1}s\}.$$

1.4 Conjugation and the class equation

A group acts on itself by *conjugation*, ${}^g x = gxg^{-1}$. The identity fixes x, and

$${}^g({}^h x) = ghxh^{-1}g^{-1} = {}^{gh}x.$$

Apply orbit–stabiliser to this action.

Definition 1.14 (Classes, centralisers, centre)**.** The orbit of g under conjugation is its *conjugacy class* $[g] = \{hgh^{-1} : h \in G\}$. The stabiliser is the *centraliser* $C_G(g) = \{h : hg = gh\}$. The *centre* is $Z(G) = \{z : zh = hz \text{ for all } h \in G\}$.

So $z \in Z(G)$ exactly when $[z] = \{z\}$, and G is abelian exactly when every class has one element. $Z(G)$ is also the kernel of conjugation, which is therefore faithful exactly when $Z(G) = \mathbf{1}$.

Theorem 1.15 (The class equation)**.** *Let G be finite.*

(i) $|[g]| = [G : C_G(g)]$*, which divides* $|G|$*.*

(ii) Let $g_1, \ldots, g_m$ be one element from each class with more than one element. Then

$$|G| = |Z(G)| + \sum_{i=1}^{m} [G : C_G(g_i)],$$

and each term of the sum divides $|G|$ and exceeds 1*.*

Proof. (i) is Theorem 1.10 for conjugation, with Corollary 1.11.

(ii) The classes partition G. The classes with one element are the $\{z\}$ with $z \in Z(G)$; the others have $[G : C_G(g_i)] > 1$ elements, by the class-size formula. ∎

Example 1.16 (The classes of S_3)**.** First,

$$(1\,3)(1\,2)(1\,3)^{-1} = (2\,3),$$
$$(2\,3)(1\,2)(2\,3)^{-1} = (1\,3).$$

So $C_{S_3}((1\,2))$ is a proper subgroup containing $\{e, (1\,2)\}$; its order divides 6 and is even (Corollary 1.11, twice), so it is 2, and the class of $(1\,2)$ is the three transpositions. Likewise $C_{S_3}((1\,2\,3))$ is proper and contains $\langle(1\,2\,3)\rangle$, of order 3; so the class of $(1\,2\,3)$ has 2 elements, the two 3-cycles. With $\{e\}$, these classes hold all six elements:

class	representative g	$\lvert[g]\rvert$	$C_{S_3}(g)$
$\{e\}$	e	1	S_3
$\{(1\,2), (1\,3), (2\,3)\}$	$(1\,2)$	3	$\{e, (1\,2)\}$
$\{(1\,2\,3), (1\,3\,2)\}$	$(1\,2\,3)$	2	$\langle(1\,2\,3)\rangle$

So $Z(S_3) = \mathbf{1}$, and the class equation reads

$$6 = 1 + 3 + 2.$$

Lecture 2 generalises this.

Conjugation moves subgroups too. The map $x \mapsto gxg^{-1}$ is an isomorphism $G \to G$, with inverse $x \mapsto g^{-1}xg$, so for $H \leq G$ the image $gHg^{-1} = \{ghg^{-1} : h \in H\}$ is a subgroup, a *conjugate* of H; and $g \cdot H = gHg^{-1}$ is an action of G on the set of its subgroups, by the computation that made conjugation an action.

Corollary 1.17. *For $H \leq G$, the stabiliser of H is its* normaliser $N_G(H) = \{g : gHg^{-1} = H\}$*, and H has $[G : N_G(H)]$ conjugates.*

This is Theorem 1.10 for the action on subgroups.

Example 1.18 (Conjugate subgroups of S_3)**.** The conjugates of $\langle(1\,2)\rangle$ are

$$\langle(1\,2)\rangle, \quad \langle(1\,3)\rangle, \quad \langle(2\,3)\rangle.$$

The normaliser therefore has index 3 and equals $\langle(1\,2)\rangle$. In contrast, $\langle(1\,2\,3)\rangle$ is its only conjugate.

Lecture 2 calls a subgroup with $N_G(H) = G$ *normal*; Lecture 3 counts Sylow subgroups by their normalisers.

1.5 Cayley and orbit counting

Theorem 1.19 (Cayley)**.** *Every group G is isomorphic to a subgroup of S_G. If $|G| = n$, it is isomorphic to a subgroup of S_n.*

Proof. Let G act on itself by left multiplication, $g \cdot x = gx$. Since $g \cdot e = g$, only e fixes e, so the kernel is $\mathbf{1}$ and the action is faithful. By Proposition 1.7, $G \cong \operatorname{im}\rho \leq S_G$. Numbering the elements of G as $1, \ldots, n$ identifies S_G with S_n. ∎

So every finite group is a group of permutations, though Cayley's embedding is seldom economical: it puts D_3 into S_6, where the vertices gave $D_3 \cong S_3$.

The second consequence of orbit–stabiliser counts orbits.

Theorem 1.20 (Orbit counting). *Let a finite group G act on a finite set X, and let* $\mathrm{Fix}(g) = \{x : g \cdot x = x\}$. *The number of orbits is*

$$\frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)| .$$

This is often called Burnside's lemma.

Proof. Count the set $P = \{(g, x) \in G \times X : g \cdot x = x\}$ in two ways. By g, $|P| = \sum_g |\mathrm{Fix}(g)|$. By x, and by Corollary 1.11,

$$|P| = \sum_{x \in X} |G_x| = \sum_{x \in X} \frac{|G|}{|G \cdot x|}.$$

Group the x by orbits: an orbit O contributes $|O|$ terms, each $|G|/|O|$, so $|G|$ in all. Hence $|P|$ is $|G|$ times the number of orbits. ∎

1.6 Prime-power groups and Cauchy's theorem

By Lagrange, the order of every element divides $|G|$. Conversely, every prime divisor of $|G|$ is the order of an element; the proof counts fixed points modulo p. For an action of G on X, let $X^G = \{x \in X : g \cdot x = x \text{ for all } g \in G\}$.

Lemma 1.21 (Fixed points modulo p). *Let p be prime, let $|G| = p^a$, and let G act on a finite set X. Then $|X| \equiv |X^G| \pmod{p}$.*

Proof. The orbits partition X. By Corollary 1.11 an orbit has $|G|/|G_x|$ elements, a divisor of p^a, so it has one element or a multiple of p. The orbits with one element are the points of X^G. ∎

Theorem 1.22 (Cauchy). *If G is finite and a prime p divides $|G|$, then G has an element of order p.*

Proof. Let X be the set of p-tuples $(g_1, \dots, g_p) \in G^p$ with $g_1 g_2 \cdots g_p = e$. The first $p-1$ entries are arbitrary, and they fix $g_p = (g_1 \cdots g_{p-1})^{-1}$; so $|X| = |G|^{p-1}$, a multiple of p. A cyclic shift keeps a tuple in X:

$$g_p g_1 \cdots g_{p-1} = g_p (g_1 \cdots g_p) g_p^{-1} = e.$$

So C_p, generated by $\zeta = \exp(2\pi \mathrm{i}/p)$, acts on X, with ζ^k shifting each tuple k places; this is well defined, since a shift by p places changes nothing. A tuple fixed by ζ has all its entries equal, so X^{C_p} consists of the $(g, \dots, g)$ with $g^p = e$. By Lemma 1.21,

$$|X^{C_p}| \equiv |X| \equiv 0 \pmod{p}.$$

It contains $(e, \dots, e)$, so it has at least $p \geq 2$ elements, and some $g \neq e$ has $g^p = e$. Then $o(g)$ divides p: writing $p = q\,o(g) + r$ with $0 \leq r < o(g)$ gives $g^r = e$, so $r = 0$. As $o(g) \neq 1$, $o(g) = p$. ∎

Example 1.23 (Cauchy's count in S_3). For $p = 3$, the set X has 36 elements. The fixed tuples are (g, g, g) with

$$g \in \{e, (1\,2\,3), (1\,3\,2)\}.$$

The fixed-point congruence reads

$$36 \equiv 3 \equiv 0 \pmod{3}.$$

Lecture 3 replaces p by its largest power dividing $|G|$: Sylow's theorems.

1.7 Worked example

In how many ways can the vertices of a triangle be coloured with k colours, if colourings that differ by a symmetry count as one?

A colouring is a map $c\colon \{1,2,3\} \to \{1,\dots,k\}$; there are k^3. The symmetries act on them, through $D_3 \cong S_3$ (Example 1.5), by

$$(g \cdot c)(i) = c(g^{-1}(i)) :$$

the colour that c put at vertex j moves to vertex $g(j)$. This is an action, since

$$\begin{aligned}(g \cdot (h \cdot c))(i) &= c(h^{-1}g^{-1}(i)) \\ &= ((gh) \cdot c)(i).\end{aligned}$$

Example 1.24 (Guard: the inverse is needed)**.** The rule $g \cdot c = c \circ g$ gives $g \cdot (h \cdot c) = c \circ h \circ g$, while $(gh) \cdot c = c \circ g \circ h$. For $k \geq 2$ these differ: with $g = (1\,2)$, $h = (1\,3)$ and c giving colour 2 to vertex 1 alone, the first gives colour 2 to vertex 3, the second to vertex 2. The dual representation (Lecture 6) and the action on tensor factors (Lecture 14) need the same inverse.

Now $g \cdot c = c$ exactly when c is constant on each cycle of g, so $|\mathrm{Fix}(g)| = k^{(\text{number of cycles of } g)}$, counting fixed points as cycles:

g	e	transpositions	3-cycles
how many	1	3	2
$\lvert\mathrm{Fix}(g)\rvert$	k^3	k^2	k

By Theorem 1.20, the number of orbits is

$$\frac{k^3 + 3k^2 + 2k}{6}.$$

For $k = 2$ the count is

$$\frac{8 + 12 + 4}{6} = 4.$$

The four orbits are



of sizes 1, 3, 3, 1. For $k = 3$ the count is

$$\frac{27 + 27 + 6}{6} = 10.$$

Self-check. How many rotations carry a cube to itself? (Use orbit–stabiliser on the faces.) *Answer:* 24. The rotations act transitively on the 6 faces, and the stabiliser of a face is the 4 rotations about the axis through its centre. Thus the group has order

$$6 \cdot 4 = 24.$$

Summary.

- *The main result:* $gG_x \mapsto g \cdot x$ identifies G/G_x with $G \cdot x$ (Theorem 1.10). It gives Lagrange (Corollary 1.11), the class equation (Theorem 1.15), orbit counting (Theorem 1.20) and Cauchy's theorem (Theorem 1.22).
- *Handed on:* actions as homomorphisms $G \to S_X$, cosets, classes, centralisers, normalisers, and $S_3 \cong D_3$.
- *Next:* G/H is not in general a group (Example 1.9); Lecture 2 makes G/N one when N is normal, and finds the classes of S_n.

References

Dummit and Foote, *Abstract Algebra*: D_n (their D_{2n}), §1.2, pp. 23–26; actions, §1.7, pp. 41–43; orbit–stabiliser, §4.1, Prop. 2, pp. 114–115; Cayley, §4.2, Cor. 4, p. 120; the class equation, §4.3, Prop. 6 and Thm. 7, pp. 123–124. Rotman, *An Introduction to the Theory of Groups*: Lagrange, Thm. 2.11, p. 26; orbit–stabiliser, Thm. 3.19, p. 57; counting orbits, Thm. 3.22, pp. 58–59; Cauchy, with McKay's proof, Thm. 4.2, p. 74. Humphreys, *A Course in Group Theory*, Ch. 10, pp. 89–94.